LOCAL COHOMOLOGY ON THE CLOSED SUBSETS DEFINED BY AN IDEAL AND ITS EXTENSION AGREE UNDER SCALAR RESTRICTION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterisation of local cohomology as a right adjoint, or its description through the alternating Cech complex. Invoked inside the counted proof (source lines 1555-1559 and 1560-1574). Gives the two routes to the identification.

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $A \to B$ be a ring homomorphism and let $I \subset A$ be a finitely generated ideal. Set $J = IB$. Set $Z = V(I)$ and $Y = V(J)$. Then*

$$R\Gamma_Z(M_A) = R\Gamma_Y(M)_A$$

functorially in $M \in D(B)$. Here $(-)_A$ denotes the restriction functors $D(B) \to D(A)$ and $D_{J^\infty\text{-}torsion}(B) \to D_{I^\infty\text{-}torsion}(A)$.

Proof. This follows from uniqueness of adjoint functors as both $R\Gamma_Z((-)_A)$ and $R\Gamma_Y(-)_A$ are right adjoint to the functor $D_{I^\infty\text{-torsion}}(A) \to D(B)$, $K \mapsto K \otimes^{\mathbf{L}}_A B$. Alternatively, one can use the description of $R\Gamma_Z$ and $R\Gamma_Y$ in terms of alternating Čech complexes (Lemma lemma-local-cohomology-adjoint). Namely, if $I =$

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$(f_1, \ldots, f_r)$ then J is generated by the images $g_1, \ldots, g_r \in B$ of $f_1, \ldots, f_r$. Then the statement of the lemma follows from the existence of a canonical isomorphism

$$\begin{aligned} & M_A \otimes_A (A \to \prod\nolimits_{i_0} A_{f_{i_0}} \to \prod\nolimits_{i_0<i_1} A_{f_{i_0}f_{i_1}} \to \ldots \to A_{f_1\ldots f_r}) \\ & = M \otimes_B (B \to \prod\nolimits_{i_0} B_{g_{i_0}} \to \prod\nolimits_{i_0<i_1} B_{g_{i_0}g_{i_1}} \to \ldots \to B_{g_1\ldots g_r}) \end{aligned}$$

for any B-module M. $\square$